1. **Double Transposition Cipher:** Use two keywords (often of different lengths). Do a transposition cipher with the first keyword to produce a first ciphertext. If there are spaces, ignore them. Then encrypt that ciphertext (as though it were plaintext) using the second keyword.

   Encryption

   i. Write the plaintext in rows.

   ii. Permute the columns using the keyword.

   iii. Read off the **columns** to form the new "plaintext"

   iv. Write it in rows in the second grid.

   v. Permute the columns using the second keyword.

   vi. Read off the **columns** to form the final ciphertext.

   Example: Encrypt the word TRANSPOSITION using the first key `KEY` and the second key `SHORT`.

K	E	Y
T	R	A
N	S	P
O	S	I
T	I	O
N		

→

E	K	Y
R	T	A
S	N	P
S	O	I
I	T	O
	N	

→ RSSIT NOTNA PIO

S	H	O	R	T
R	S	S	I	T
N	O	T	N	A
P	I	O		

→

H	O	R	S	T
S	S	I	R	T
O	T	N	N	A
I	O		P	

→ SOIST OINRN PTA

Decryption:

i. Count the number of characters. Determine how many "extra" characters there will be for the second keyword.

ii. Write your second grid with the permuted keyword.

iii. Fill in the **columns** of the grid, making sure to put the extra characters in the appropriate columns. Mark those columns with *. For example, if your second keyword is `SHORT` and you have two "extra" characters, then columns S and H get one more letter than the other columns.

iv. (Un)permute the columns.

v. Read off the **rows**.

vi. Using the number of characters, determine the number of "extra" characters for the first keyword.

vii. Fill in the columns of the grid, filling in the extra long columns (marked with *) using the sorted keyword. For example, if your first keyword is `KEY` and you have two "extra" characters, then columns K and E get one more letter than the other columns.

viii. Unpermute the columns.

ix. Read off the **rows**

Decrypt what we got when we encrypted TRANSPOSITION.

SOI|ST OI|N|RN P|TA

① 13 characters
5 columns

$\frac{13}{5} = 2 \text{ R } 3$ ← 3 special columns

* get 3 chara, else 2

* H	* O	R	* S	T
S	S	I	R	T
O	T	N	N	A
I	O		P	

→

* S	* H	* O	R	T
R	S	S	I	T
N	O	T	N	A
P	I	O		

↦

E	* K	Y
R	T	A
S	N	P
S	O	I
I	T	O
	N	

→

* K	E	Y
T	R	A
N	S	P
O	S	I
T	I	O
N		

✓

② $\frac{13}{3} = 4 \text{ R } 1$

Decrypt the ciphertext E T|E T|S P|I H X|C R assuming it was encrypted using the scheme above.

$11/5 = 2 \text{ R } 1$

H	O	R	* S	T
E	E	S	I	C
T	T	P	H	R
			X	

→

* S	H	O	R	T
I	E	E	S	C
H	T	T	P	R
X				

→

$11/3 = 3 \text{ R } 2$

* E	* K	Y
I	C	P
E	H	R
E	T	X
S	T	

→

* K	* E	Y
C	I	P
H	E	R
T	E	X
T	S	

CIPHERTEXTS